Lecture 14 (Revised Simplex Method and Simplex Tableau)

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We will improve our simplex method's runtime by deriving two modifications of the simplex method.

Revised Simplex Method

Currently the biggest bottleneck in computation for the simplex method is finding B^{-1}. This process is $O\left(m^3\right)$. Can we speed this up? Yes! we will do so by using the **Revised Simplex Method**. The key idea is to use Elementary Row Operations (EROs) in order to speed up this process. Let

$$B = \begin{bmatrix} A_{B(1)} & A_{B(2)} & \dots & A_{B(m)} \end{bmatrix}$$

be the current basis matrix and let

$$\bar{B} = \begin{bmatrix} A_{B(1)} & \dots & A_{B(l-1)} & A_j & A_{B(l+1)} & \dots & A_{B(m)} \end{bmatrix}$$

be the new basis matrix. Then note that

$$B^{-1}B = I$$

and if we look entry-wise, we see

$$B^{-1}\bar{B} = \begin{bmatrix} 1 & 0 & \dots & u_1 & \dots & 0 \\ 0 & 1 & \dots & u_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & u_l & \dots & 0 \\ \vdots & \vdots & \dots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & u_m & \dots & 1 \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & \dots & e_{l-1} & \vec{u} & e_l & \dots & e_m \end{bmatrix}$$

Elementary row operations can be applied by applying left multiplication by an invertible matrix, Q_i. A sequence of EROs can be applied by multiplying by a sequence of these ERO matrices, $Q = Q_m Q_{m-1} \dots Q_2 Q_1$. Suppose we apply the following EROs to $B^{-1}\bar{B}$ (With the bottom ERO applied last):

$$R_i - \frac{u_i}{u_l} R_l \to R_i \; i \neq l$$

$$\frac{1}{u_l} R_l \to R_l$$

Then $B^{-1}\bar{B}$ will turn into the identity matrix. If we represent this sequence of EROs as Q, then we see

$$QB^{-1}\bar{B} = I$$

Thus

$$\bar{B}^{-1} = QB^{-1}$$

in other words, we can apply these EROs to B^{-1} and get $\bar{B}^{-1}$ for much cheaper than a naive inverse algorithm. In fact, this sequence of EROs can be computed in $O\left(m^2\right)$.

Example

Let $B^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 1 \\ 4 & -3 & -2 \end{pmatrix}$, $\vec{u} = \begin{pmatrix} -4 \\ 2 \\ 2 \end{pmatrix}$, and $l = 3$. Then we want to turn $\vec{u}$ into e_3. To do so, we will apply the following sequence of EROs in order:

$$R_1 - \frac{u_1}{u_3} R_3 \to R_1$$

$$R_2 - \frac{u_2}{u_3} R_3 \to R_2$$

$$\frac{1}{u_3} R_3 \to R_3$$

Substituting in the values of $\vec{u}$, this yields

$$R_1 - \frac{-4}{2} R_3 \to R_1$$

$$R_2 - \frac{2}{2} R_3 \to R_2$$

$$\frac{1}{2} R_3 \to R_3$$

Simplify:

$$R_1 + 2R_3 \to R_1$$

$$R_2 - R_3 \to R_2$$

$$\frac{1}{2} R_3 \to R_3$$

Applying these EROs in order yields the following matrices:

$$B^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 1 \\ 4 & -3 & -2 \end{pmatrix} \to \begin{pmatrix} 9 & -4 & -1 \\ -2 & 3 & 1 \\ 4 & -3 & -2 \end{pmatrix} \to \begin{pmatrix} 9 & -4 & -1 \\ -6 & 6 & 3 \\ 4 & -3 & -2 \end{pmatrix} \to \begin{pmatrix} 9 & -4 & -1 \\ -6 & 6 & 3 \\ 2 & -\frac{3}{2} & -1 \end{pmatrix} = \bar{B}^{-1}$$

This yields the Revised Simplex Algorithm.

1) Start with the basis, $A_{B(1)}, \ldots, A_{B(m)}$, the BFS, $\vec{x}$, and B^{-1} for the basis matrix, B.

2) Compute the row vector, $\vec{p}^T = \vec{c}_B^T B^{-1}$ and the reduced costs $\bar{c}_j = c_j - \vec{p}^T A_j$. If all of the reduced costs are non-negative, then the current BFS is optimal and the algorithm terminates. Otherwise, pick j such that $\bar{c}_j < 0$.

3) Compute $\vec{u} = B^{-1} A_j$. If no component of $\vec{u}$ is positive, then the optimal cost is $-\infty$, and the LP is unbounded. In this case, terminate.

4) If some component of $\vec{u}$ is positive, let

$$\theta^* = \min_{\{i=1,\ldots,m | u_i > 0\}} \frac{x_{B(i)}}{u_i}$$

5) Let l be such that $\theta^* = \frac{x_{B(l)}}{u_l}$. Form a new basis by replacing $A_{B(l)}$ with A_j. If $\vec{y}$ is the new BFS, then $y_j = \theta^*$ and $y_{B(i)} = x_{B(i)} - \theta^* u_i$ for $i \neq l$.

6) Form $\left[B^{-1} | \vec{u}\right]$. Turn the last column into e_l by EROs. After applying these EROs, the first m columns will form $\bar{B}^{-1}$.

Simplex Tableau

Instead of storing a few matrices, we can store all of the relevant information in one large block matrix called the **Simplex Tableau**. The idea is that instead of updating B^{-1} directly, we can update

$$B^{-1}\left[\vec{b}|A\right]$$

This augmented matrix has columns, $B^{-1}\vec{b}$ (we call this the 0th column) and has columns $B^{-1}A_1, B^{-1}A_2, \ldots, B^{-1}A_n$ (we call these columns 1 through n). At the end of each iteration of the algorithm, we update the augmented matrix to get

$$\bar{B}^{-1}\left[\vec{b}|A\right]$$

This is done by left multiplying by Q (as we did with the revised simplex method). The key idea is still to apply EROs to this augmented matrix. Typically we will also add one row at the top of the augmented matrix (we call this the 0th row). The full augmented matrix (with the 0th row) is called the Simplex Tableau. This is of the form

$$\left[\begin{array}{c|c} -\vec{c}_B^T B^{-1}\vec{b} & \vec{c} - \vec{c}_B^T B^{-1}A \\ - & - \\ B^{-1}\vec{b} & B^{-1}A \end{array}\right]$$

Notice the top left corner contains the negative of the cost, the bottom left corner contains the entries of $\vec{x}_B$, the top right corner contains the reduced costs, and the bottom right corner contains all the $B^{-1}A_i$. In other words, our matrix looks like:

$$\left[\begin{array}{ccccc} -\vec{c}_B^T\vec{x}_B & \bar{c}_1 & \bar{c}_2 & \ldots & \bar{c}_n \\ x_{B(1)} & | & | & \ldots & | \\ \vdots & B^{-1}A_1 & B^{-1}A_2 & \ldots & B^{-1}A_n \\ x_{B(m)} & | & | & \ldots & | \end{array}\right]$$

It turns out if we update all of the rows (including the 0th row!) by row reduction, then we are able to apply a step of the simplex method appropriately. Note the 0th row looks like

$$\left[0|\vec{c}^T\right] - \vec{c}_B^T B^{-1}\left[\vec{b}|A\right]$$

and after updating, we get

$$\left[0|\vec{c}^T\right] - \vec{c}_B^T \bar{B}^{-1}\left[\vec{b}|A\right]$$

This gives the full Simplex Tableau Algorithm:

1) Start with the basis matrix, B and BFS, $\vec{x}$.

2) Examine the reduced costs in the 0th row. If all of the costs are non-negative, then the current BFS is optimal and we terminate. Otherwise, we pick j such that $\bar{c}_j < 0$.

3) Consider $\vec{u} = B^{-1}A_j$. If no component of $\vec{u}$ is positive, then the optimal cost is $-\infty$ and the LP is unbounded. In this case, terminate the algorithm.

4) For all i where $u_i > 0$, compute $\frac{x_{B(i)}}{u_i}$. Let l be the index of the row with the smallest ratio. Then $A_{B(l)}$ exists the basis and A_j enters.

5) In each row, add a constant multiple of the lth row so that u_l becomes 1 and all other entries in the jth column become 0.

Example

Consider the following LP.

$$\text{minimize } -10x_1 - 12x_2 - 12x_3$$

subject to

$$x_1 + 2x_2 + 2x_3 \leq 20$$

$$2x_1 + x_2 + 2x_3 \leq 20$$

$$2x_1 + 2x_2 + x_3 \leq 20$$

$$x_1, x_2, x_3 \geq 0$$

This LP isn't in standard form. The first step is to convert it to standard form. We will add some slack variables.

$$\text{minimize } -10x_1 - 12x_2 - 12x_3$$

subject to

$$x_1 + 2x_2 + 2x_3 + x_4 = 20$$

$$2x_1 + x_2 + 2x_3 + x_5 = 20$$

$$2x_1 + 2x_2 + x_3 + x_6 = 20$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

Now, we note that $\vec{x} = \begin{pmatrix} 0 & 0 & 0 & 20 & 20 & 20 \end{pmatrix}$ is a BFS. Suppose we use this as our starting BFS. Then this corresponds to $B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, and $B(1) = 4$, $B(2) = 5$, and $B(3) = 6$. As such, we see $B^{-1} = B = I$. We can now fill in the starting simplex tableau.

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ & 0 & -10 & -12 & -12 & 0 & 0 & 0 \\ x_4 = & 20 & 1 & 2 & 2 & 1 & 0 & 0 \\ x_5 = & 20 & 2 & 1 & 2 & 0 & 1 & 0 \\ x_6 = & 20 & 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(Here, we added the left entries in a -1st column to keep the active variables straight. Technically this isn't needed). Notice that x_1, x_2, x_3 all have negative reduced costs. We select one of these to enter the basis. Suppose we pick x_1. Then this yields $\vec{u} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. Then each entry of $\vec{u}$ is positive. We compute the ratios

$$\frac{x_{B(i)}}{u_i} \to \left\{ \frac{20}{1}, \frac{20}{2}, \frac{20}{2} \right\} = \{20, 10, 10\}$$

Notice for $l = 2, 3$ we have the smallest positive ratio. Let's select $l = 2$. Then we select $B(2) = x_5$ to leave the basis and $j = 1$ as the index to enter the basis. This means we should have the new basis $B(1) = 4, B(2) = 1, B(3) = 6$. We will take u_2 to be the pivot and we will row reduce the simplex tableau.

$$\begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 0 & -10 & -12 & -12 & 0 & 0 & 0 \\ 20 & 1 & 2 & 2 & 1 & 0 & 0 \\ 20 & 2 & 1 & 2 & 0 & 1 & 0 \\ 20 & 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_0 + 5R_2 \to R_0} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 100 & 0 & -7 & -2 & 0 & 5 & 0 \\ 20 & 1 & 2 & 2 & 1 & 0 & 0 \\ 20 & 2 & 1 & 2 & 0 & 1 & 0 \\ 20 & 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1 - \frac{1}{2}R_2 \to R_1} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 100 & 0 & -7 & -2 & 0 & 5 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 20 & 2 & 1 & 2 & 0 & 1 & 0 \\ 20 & 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2 \to R_3} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 100 & 0 & -7 & -2 & 0 & 5 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 20 & 2 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2 \to R_2} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 100 & 0 & -7 & -2 & 0 & 5 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 10 & 1 & 0.5 & 1 & 0 & 0.5 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{bmatrix}$$

Or with labels:

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ & 100 & 0 & -7 & -2 & 0 & 5 & 0 \\ x_4 = & 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ x_1 = & 10 & 1 & 0.5 & 1 & 0 & 0.5 & 0 \\ x_6 = & 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{bmatrix}$$

Our new BFS is $\vec{x} = \begin{pmatrix} 10 & 0 & 0 & 10 & 0 & 0 \end{pmatrix}$. This is a degenerate BFS. We completed the first iteration. Let's do another. Notice that x_2 and x_3 both have negative reduced costs. Suppose we select x_3 to activate. Then $\vec{u} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$. We now can check each of the ratios.

$$\frac{x_{B(i)}}{u_i} \to \left\{ \frac{10}{1}, \frac{10}{1}, \frac{0}{-1} \right\} = \{10, 10, 0\}$$

(Note we do not need to compute the last one since $u_3 < 0$). In this case, both $l = 1, 2$ are minimizing. Suppose we take $l = 1$. Then x_4 will exit the basis and x_3 will enter the basis. We select row 1, column 3 to be our pivot, and we row reduce.

$$\begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 100 & 0 & -7 & -2 & 0 & 5 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 10 & 1 & 0.5 & 1 & 0 & 0.5 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_0 + 2R_1 \to R_0} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 120 & 0 & -4 & 0 & 2 & 4 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 10 & 1 & 0.5 & 1 & 0 & 0.5 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_2 - R_1 \to R_2} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 120 & 0 & -4 & 0 & 2 & 4 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 0 & 1 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 + R_1 \to R_3} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 120 & 0 & -4 & 0 & 2 & 4 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 0 & 1 & -1 & 0 & -1 & 1 & 0 \\ 10 & 0 & 2.5 & 0 & 1 & -1.5 & 1 \end{bmatrix}$$

Or with labels:

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ & 120 & 0 & -4 & 0 & 2 & 4 & 0 \\ x_3 = & 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ x_1 = & 0 & 1 & -1 & 0 & -1 & 1 & 0 \\ x_6 = & 10 & 0 & 2.5 & 0 & 1 & -1.5 & 1 \end{bmatrix}$$

This yields the new BFS of $\vec{x} = \begin{pmatrix} 0 & 0 & 10 & 0 & 0 & 10 \end{pmatrix}$. Let's do the last iteration. Since x_2 has a negative reduced cost, we select x_2 to activate. This yields $\vec{u} = \begin{pmatrix} 1.5 \\ -1 \\ 2.5 \end{pmatrix}$, so we get the ratios

$$\frac{x_{B(i)}}{u_i} \to \left\{ \frac{10}{1.5}, \frac{0}{-1}, \frac{10}{2.5} \right\} = \{15, 0, 4\}$$

so we take $l = 3$ and have $B(3) = x_6$ exit the basis with x_2 entering the basis. This means we take row 3, column 2 to be the pivot. Thus

$$\begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 120 & 0 & -4 & 0 & 2 & 4 & 0 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 0 & 1 & -1 & 0 & -1 & 1 & 0 \\ 10 & 0 & 2.5 & 0 & 1 & -1.5 & 1 \end{bmatrix} \xrightarrow{R_0 + \frac{4}{2.5} R_3 \to R_0} \begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 136 & 0 & 0 & 0 & 3.6 & 1.6 & 1.6 \\ 10 & 0 & 1.5 & 1 & 1 & -0.5 & 0 \\ 0 & 1 & -1 & 0 & -1 & 1 & 0 \\ 10 & 0 & 2.5 & 0 & 1 & -1.5 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1-\frac{1.5}{2.5}R_3\to R_1}\begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 136 & 0 & 0 & 0 & 3.6 & 1.6 & 1.6 \\ 4 & 0 & 0 & 1 & 0.4 & 0.4 & -0.6 \\ 0 & 1 & -1 & 0 & -1 & 1 & 0 \\ 10 & 0 & 2.5 & 0 & 1 & -1.5 & 1 \end{bmatrix}$$

$$\xrightarrow{R_2+\frac{1}{2.5}R_3\to R_2}\begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 136 & 0 & 0 & 0 & 3.6 & 1.6 & 1.6 \\ 4 & 0 & 0 & 1 & 0.4 & 0.4 & -0.6 \\ 4 & 1 & 0 & 0 & -0.6 & 0.4 & 0.4 \\ 10 & 0 & 2.5 & 0 & 1 & -1.5 & 1 \end{bmatrix}\xrightarrow{\frac{1}{2.5}R_3\to R_3}\begin{bmatrix} * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ 136 & 0 & 0 & 0 & 3.6 & 1.6 & 1.6 \\ 4 & 0 & 0 & 1 & 0.4 & 0.4 & -0.6 \\ 4 & 1 & 0 & 0 & -0.6 & 0.4 & 0.4 \\ 4 & 0 & 1 & 0 & 0.4 & -0.6 & 0.4 \end{bmatrix}$$

Or with labels:

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ & 136 & 0 & 0 & 0 & 3.6 & 1.6 & 1.6 \\ x_3 = & 4 & 0 & 0 & 1 & 0.4 & 0.4 & -0.6 \\ x_1 = & 4 & 1 & 0 & 0 & -0.6 & 0.4 & 0.4 \\ x_2 = & 4 & 0 & 1 & 0 & 0.4 & -0.6 & 0.4 \end{bmatrix}$$

Thus $\vec{x} = \begin{pmatrix} 4 & 4 & 4 & 0 & 0 & 0 \end{pmatrix}$ is the new BFS. As there are no longer any negative reduced costs, then we see this BFS is optimal. Further, we see the cost of the objective function is -136 at this optimal solution. As a final thought, it is usually convenient to use Excel to track each tableau. Writing these by hand is rather tedious. Feel free to use a spreadsheet like Excel to track these (as long as you attach screenshots of your work as you use a spreadsheet). Many spreadsheets can help automate calculations. Later, we will use Python to automate these calculations too.